CSC D18 – Computer Graphics – A3 features checklist

For people working in teams, we only need one name + student number to find your submitted work

Student Name (last, first): Mao, Fengxi (Brook) Student No.: 1010146815 UTORid: maofengx

Place an ***X*** in the corresponding column for each features

Feature	*Value*	*Working*	*Partial*	*Not done*	*Test image name*
Antialiasing	2.5	**X**			antialiasing.ppm
Texture Mapping	5	**X**			texture_mapping.ppm
ALS and soft shadows	5	**X**			area_light.ppm
Cool scene	up to 15	**X**			fibonacci_word.ppm
Refraction	up to 10	**X**			refraction.ppm
Multi-threading	2	**X**			(none, see note)
CSG or Hierarchical	Up to 10		**X**		fibonacci_word.ppm
Normal mapping	5	**X**			normal_mapping.ppm
Alpha or other mappings	2.5	**X**			alpha_mapping.ppm
Depth of field	up to 7				
Photon mapping	up to 12				
Dispersion	5				
RT Acceleration	up to 12	**X**			(none, see note)
Ray Marching	up to 7				

Provide a list of scripts that must be run to generate the test images noted above:

Just run demo/scripts/run_all.sh (works from any directory). It runs all of the scripts below (they're all in demo/scripts/) at the same time. Each one compiles its own binary into demo/bin/ with some -D flags (they're listed at the top of buildscene.c) and renders a 1024x1024 image into demo/renderings/.

```
base.sh             base.ppm             no A3 features, just for comparison
antialiasing.sh     antialiasing.ppm     base scene with antialiasing on
texture_mapping.sh  texture_mapping.ppm  -DTEXTURES
normal_mapping.sh   normal_mapping.ppm   -DTEXTURES -DNORMALS
area_light.sh       area_light.ppm       -DALS
alpha_mapping.sh    alpha_mapping.ppm    -DALPHA
refraction.sh       refraction.ppm       -DSCENE_REFRACTION (a refractive sphere in another one)
fibonacci_word.sh   fibonacci_word.ppm   -DSCENE_FIBONACCI (the cool scene)
```

The cool scene

It's meant to look like this photo of OCAD U, except the random black checker-like pattern on the bottom is swapped out for a Fibonacci word fractal (youtu.be/g3ciepKkWQI):

`https://admissions.ocadu.ca/sites/default/files/2025-08/54507200903_ebee7070ba_k%20%281%29.jpg`

Multi-threading and RT acceleration

There's no real way to show these off in an image, so you'll have to take my word for it:

- Multi-threading: with a thread pool, a scene that took 10-11s to render went down to 2-3s.
- RT acceleration: with a BVH tree, a scene that took 218s to render went down to 6s.